AFFECTIVE COMPUTING & MACHINE VISION | ORIGINAL RESEARCH

An Automated and Real-time Approach of Depression Detection from Facial Micro-expressions

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Clinical Need & Objective

- Traditional challenges: Diagnosis relies on subjective clinical interviews and forms (DSM-5, PHQ-9) which suffer from patient reluctance and clinician bias.
- The Objective: Propose a non-invasive, privacy-preserving, real-time computational framework extracting subtle micro-expressions using lightweight CNN + SVM.



AFFECTIVE COMPUTING & MACHINE VISION | SYSTEM DESIGN

Dual-Track Methodology & Custom CNN Architecture

TOP: MULTI-STAGE PROCESSING & DUAL-TRACK PIPELINE

1. Video Input & Decimation

Raw Video: 24 fps, 3-5 min per patient.

Decimation: Drops 1 of 2 consecutive frames to cut compute overhead while preserving temporal resolution.

2. Preprocessing & Alignment

Toolkit: OpenFace pipeline.

Operations: Face cropping, 3D/2D landmark alignment, and intensity normalization into **256×256 RGB**.

3A. Handcrafted ML Track

Features (495): 35 Action Units, 136 Landmarks, 288 Gaze vectors, 6 Head Pose parameters.

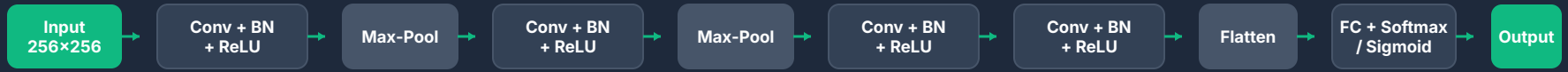
Classifier: Fed into Support Vector Machine (SVM).

3B. Deep Learning Track

Direct Input: Aligned 256×256 RGB face frames bypass traditional feature extraction.

Target: Streams directly to the custom 16-layer lightweight CNN below.

BOTTOM: 16-LAYER CUSTOM LIGHTWEIGHT CNN LAYER STACK



TECHNICAL PARAMETER CALLOUTS

Total Layers: 16 Layers Sequential Pipeline

Trainable Parameters: Only 6.5 Million (highly optimized vs. 138M in standard VGG16 models, facilitating real-time edge deployment)

HYPERPARAMETERS & TRAINING CONFIGURATION

Learning Rate: 0.001 | **Batch Size:** 128

Momentum: 0.9 | **Dropout Rate:** 0.5 (applied to FC layers)

AFFECTIVE COMPUTING & MACHINE VISION | CLINICAL PIPELINE

Clinical Cohort & Experimental Setup

TOP: CLINICAL ACQUISITION & PROTOCOL WORKFLOW

1. Patient Presentation
Bahawal Victoria Hospital

→

2. Informed Consent
Ethical clearance and protocols

→

3. Clinical Interview
Structured: PHQ-8 / PHQ-9 / DSM-V

→

4. Video Capture
3-5 min raw video at 24 fps

→

5. Frame Extraction
256×256 image generation

Clinical Cohort Breakdown

Total Participants: **380 subjects**

- **Clinically Depressed:** 180 patients (120 females, 60 males across ages 20 to 70)
- **Healthy Controls:** 200 individuals (130 males, 70 females across ages 20 to 65)

Clinical Validation Protocol:
Multi-psychiatrist evaluation based on standard PHQ-8 score thresholds and DSM-V diagnostic criteria.

Demographic Diversity:
Broad socioeconomic coverage spanning job holders, active housewives, married, and unmarried individuals.

Image & Frame Quantities

Full Raw Corpus: \~2,188,800 total video frames
Detail: 1,036,800 depressed; 1,152,000 normal frames.

Subsampled Experimental Dataset: 109,440 frames
Detail: 51,840 depressed, 57,600 normal frames selected from 38 representative subjects.

Train / Validation / Test Split:
70% Training | 15% Validation | 15% Testing

Frame Resolution: Standardized **256×256 RGB** color frames.

AFFECTIVE COMPUTING & MACHINE VISION | BENCHMARKS

Experimental Evaluation & Benchmark Comparison

PROPOSED CUSTOM CNN

99.9% Val Accuracy

Highly optimized with 6.5M parameters (Real-time capable)

HANDCRAFTED SVM TRACK

100.0% Acc & 1.00 AUC

Fusing 465 Features (AU + Landmark + Head Pose + Gaze)

NOISE SENSITIVITY WARNING

Over-tuned on Unaligned Frames

Highlights the absolute criticality of landmark-based face alignment

Feature Ablation Analysis (SVM Accuracies)

FEATURE SUBSET (COUNT)	SVM VALIDATION ACCURACY
Pose Features (6)	95.4%
Gaze Features (288)	82.1%
Action Units (35)	90.0%
Landmark Features (136)	98.5%
Combined Feature Set (465)	**100.0%**

Benchmarking Against State-of-the-Art Models



AFFECTIVE COMPUTING & MACHINE VISION | INSIGHTS

Major Findings & Clinical Takeaways

LIGHTWEIGHT EFFICIENCY

EDGE DEPLOYMENT

High Accuracy with Low Parameter Footprint

The custom CNN achieves **99.9% accuracy** with only **6.5M parameters**—a ~95% reduction in model size compared to VGG16/19 (138M–144M)—enabling real-time depression detection on resource-constrained clinical hardware and standard digital cameras.

FACIAL ALIGNMENT CRITICALITY

DATA PREPROCESSING

Raw Frames Lead to Over-tuning

Training directly on unaligned, noisy raw frames caused models to **over-tune** and memorize background artifacts. Preprocessing with **OpenFace landmark alignment** proved non-negotiable for reliable clinical generalization.

MULTI-FEATURE SYNERGY

FEATURE ENGINEERING

Landmarks and AUs Drive Superior Discrimination

While individual modalities like gaze (82.1%) or head pose (95.4%) provided partial indicators, integrating micro-expression AUs with geometric landmarks achieved **perfect 100% classification** on SVM, proving micro-expressions carry vital clinical diagnostic cues.

SCALABLE SCREENING

HEALTHCARE IMPACT

Overcoming Diagnostic Hesitation & Stigma

Real-time automated micro-expression scanning provides a **private, objective screening tool** that bypasses patient refusal and self-report distortion in socioeconomically constrained populations, paving the way for stigma-free triage.